It judges flood risk on the device

UT San Antonio’s prototype runs its AI on the device, with no server. Its release says risk at San Antonio’s low-water crossings often climbs before regional alerts sound. Drawn to illustrate.



c3 c5 c16 c19

texasaidocket.com

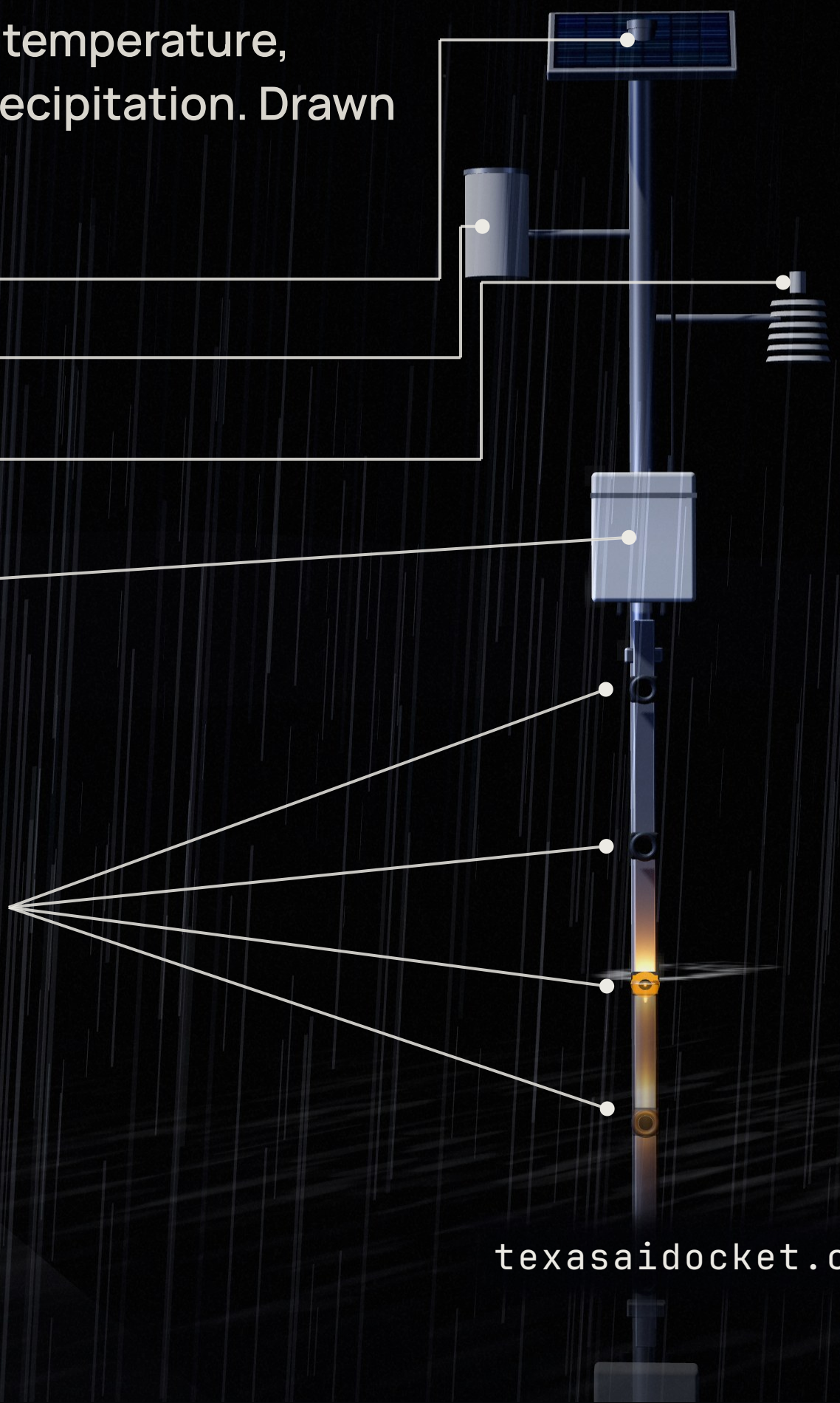
Radar can read the storm low

UT San Antonio’s release says regional models can miss very local flash floods. UT Arlington checked radar against more than 300 rain gauges and found it can read storm totals roughly 20% low under some conditions.



Four water sensors stacked on one node

Four optical water-level sensors at varying heights, plus temperature, humidity, light and precipitation. Drawn to illustrate.



c1 c2 c4 c8

texasaidocket.com

No server in the loop

TinyML shrinks a machine learning model to run on a small, low-power microcontroller. The model on the node judges imminent flood risk from current and recent readings without a central server.



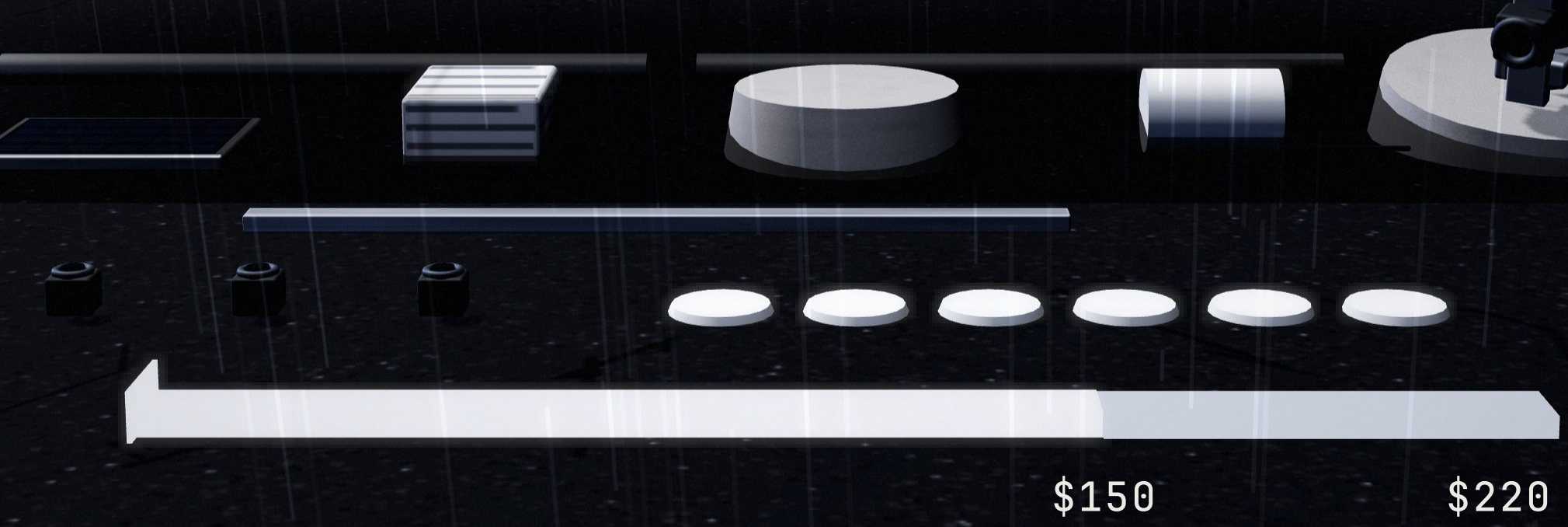
Over a half-mile in town, up to five miles open

The node’s LoRa radio reaches more than a half-mile through city streets and up to five miles in open areas, the release says. No site is named, so the circles show scale only.



$150 to $220 in parts

Parts for one prototype station cost roughly $150 to $220, the release says. A Texas Coastal Management Program grant approved by the Texas Land Commissioner helped fund the work. The parts are drawn to illustrate.



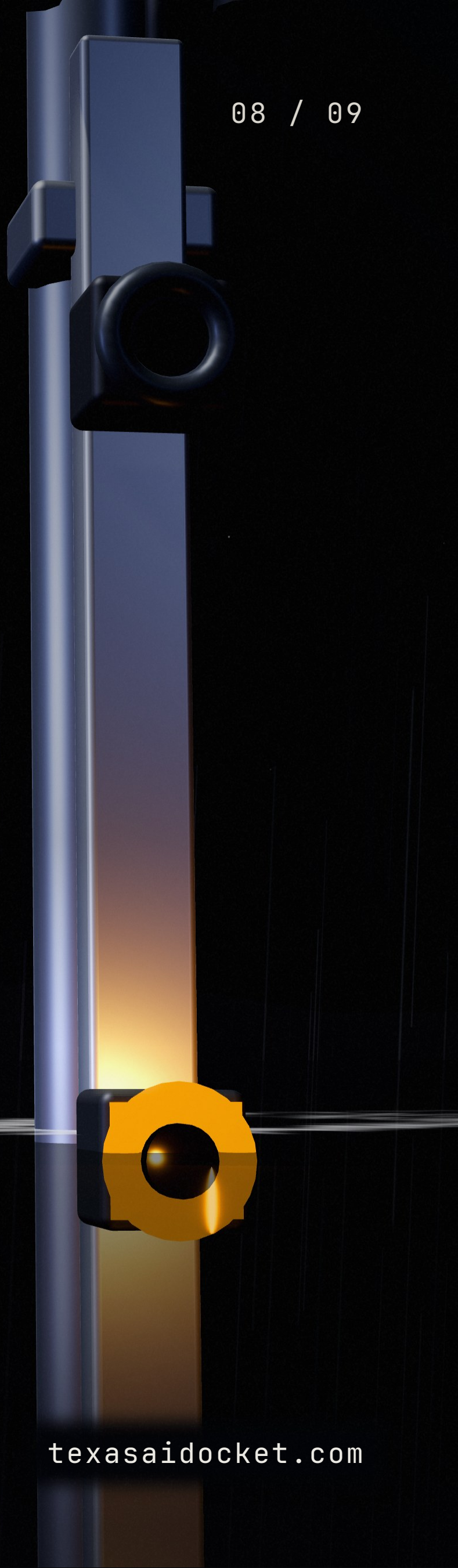
98.82%, scored on data

The release reports 98.82% validation accuracy in initial assessments after training.



No flood test is named

The release names no installation site, no city or county running it and no field test against a real flood. It gives no false alarm or missed event figure. It describes a prototype the team aims at small coastal towns. Drawn to illustrate.



Turn around at the water

The National Weather Service cites the CDC. Over half of flood drownings happen when a vehicle is driven into flood water. Whatever a sensor reports, the turn is the driver's. Drawn to illustrate.

